Robotic Awareness Event Edition 2: Multi-Robot Collaborative Operations Challenge

Summary of work completed: SUMMIT XL, Spot, Go2W, and Franka Research 3

9 October 2026

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1. Overview

The Robotic Awareness Event – Edition 2 features a *Multi-Robot Collaborative Operations Challenge* designed to demonstrate coordinated robotic operations, human–robot interaction, autonomous navigation, perception, and robotic manipulation. Participating teams execute color-coded package-delivery tasks within a time-constrained environment. Multiple robotic platforms operate in designated zones, while human operators (P1-P5) coordinate the exchange of information and physical packages between stations. Tasks are assigned different levels of complexity and corresponding point values (3, 6, or 12 points), requiring the handling of one, two, or three packages, respectively. Higher-value tasks involve additional robotic operations, package transfers, and coordination among the participating systems and operators.

This technical report documents the development, implementation, integration, and testing activities undertaken between June and October 2026 in preparation for the challenge. The work covers four robotic platforms: the SUMMIT XL wheeled mobile robot, the Boston Dynamics Spot quadruped, the Unitree Go2W wheeled-legged robot (developed as a backup platform for the SUMMIT XL), and the Franka Research 3 (FR3) robotic manipulator deployed at the central manipulation station.

The main technical developments and deliverables completed during the reporting period include:

- **Process modeling and workflow coordination:** Development and iterative refinement of a Business Process Model and Notation (BPMN) representation of the complete challenge workflow, incorporating task sequences, robot responsibilities, human–robot interactions, and package transfers between stations. The model was updated to reflect technical discussions and decisions made during project coordination meetings.
- **Simulation environment:** Development of a MuJoCo-based simulation environment representing the challenge arena and associated work-order workflows, supporting system development, operational planning, and preliminary validation.
- **Mobile robot software development:** Development and implementation of navigation and control software for the SUMMIT XL, Spot, and Go2W platforms, incorporating voice-command and clicker-based interaction, typed-command fallback, safety-related control mechanisms, and operator interfaces for monitoring and controlling robot operations.
- **Spot perception and autonomous task execution:** Implementation of QR-code scanning and package-code matching capabilities, enabling the identification and verification of packages and supporting autonomous navigation between the designated handoff and storage locations.
- **SUMMIT XL navigation improvements:** Configuration, troubleshooting, and optimization of the SUMMIT XL navigation system, including localization, navigation parameters, and robot footprint adjustments. These activities enabled reliable navigation during internal testing between the four designated challenge locations: order counter, handoff station, robotic arm station, and delivery station.
- **FR3 robotic manipulation system:** Development and integration of a perception-guided pick-and-place system using a wrist-mounted RGB-D camera, hand–eye calibration, object localization, and learned Cartesian motion trajectories based on Dynamic Movement Primitives (DMPs). The system supports automated object grasping, transfer, and placement operations.
- **FR3 operator interface:** Development of a graphical user interface (GUI) supporting system configuration, demonstration recording and teaching, motion validation, and execution of the robotic manipulation workflow, facilitating integration and testing on the physical FR3 platform.
- **Hardware modifications and fabrication:** Design, fabrication, and installation of customized trays for the SUMMIT XL and Spot platforms, together with laser-cut item

boxes, QR-coded task cards, and identification labels. These modifications and materials support physical package handling, transport, identification, and execution of the challenge tasks.

- **System integration and internal validation:** Iterative integration, troubleshooting, and internal testing of the four robotic platforms, including navigation, perception, human–robot interaction, manipulation, and task execution functionalities. Testing activities were conducted to assess platform readiness, identify technical limitations, and support progressive improvements ahead of the event.

Collectively, these activities represent the technical progress achieved during the reporting period and provide the basis for documenting the work performed, resources utilized, and deliverables completed in support of the project's objectives and associated payment milestones.

2. Tasks completed

Table 1 summarizes the main work areas and associated activities, which are described in greater detail in the following sections. Table 2 presents the key project milestones and corresponding dates.

Table 1: Summary of key elements related to the challenge

Area	What was done	Section
Process design	BPMN workflow with nine swimlanes covering the timed loop, task-level choice, Spot trips, arm handoff and delivery; revised after team meetings.	3
Simulation	MuJoCo arena with both zones, operator models, SUMMIT, Spot and arm; automatic run of three work orders.	4
Common software	Clicker-triggered voice pipeline (Whisper), closed-vocabulary command matcher, browser status display, control panel and departure hold.	5.1
SUMMIT XL	ROS and Docker setup, mapping and cropping, navigation tuning, voice/clicker bridge, status display, hold panel.	5.2
Spot	GraphNav arena map, localisation workflow, typed and voice navigation, arrival QR scanning and matching, audio and visual feedback, control panel.	5.3
Go2W	Nav2 waypoint navigation by voice and clicker, status display and control panel (backup for the SUMMIT).	5.4
Franka Research 3	Wrist-camera perception, hand-eye calibration, hand-guided pick/place recording, Cartesian DMP learning and adaptation, grasp control, MoveIt IK validation, simulation and operator GUI.	5.5
Hardware and materials	Trays on SUMMIT and Spot, item boxes, QR task cards and labels.	6
Testing	Automated tests, test modes that need no robot, and internal robot testing.	7

Work phases and timeline

Table 2: Summary of technical development activities and key milestones (June–October 2026

Phase	Main Activities and Technical Milestones
June 2026	Initial system development and workflow modeling. Developed the initial MuJoCo simulation of the challenge arena, incorporating human operator models and the SUMMIT XL robotic platform. Prepared the first BPMN model representing the overall challenge workflow and interactions between robotic platforms and operators. Initiated development of the SUMMIT XL navigation and control system, including ROS and Docker environment configuration, GPU and RViz setup, text-based navigation commands, preliminary voice-command interruption functionality, and implementation of an emergency braking mechanism using the space bar.
July 2026	Navigation, human–robot interaction, and perception development. Developed voice-command navigation and safety-handling functionalities for Spot, including map loading and localization. Implemented QR-code scanning triggered upon arrival at designated waypoints. Integrated Whisper-based speech recognition into the SUMMIT XL voice-command and clicker-control pipeline. Identified and resolved a navigation-planning issue associated with movable obstacles, improving the robot's ability to generate valid navigation paths.
August 2026	Navigation optimization, platform integration, and operator interfaces. Conducted extensive SUMMIT XL navigation troubleshooting and parameter tuning, including improvements to odometry, localization, and costmap configurations. Adapted the robot's navigation footprint and LiDAR filtering to accommodate the newly installed rear-mounted tray. Developed robot status displays for SUMMIT XL and Spot. Extended Spot's operational capabilities through audio feedback, QR-code matching, input locking to prevent conflicting commands, and clicker-based interaction. Continued software integration, internal testing, and technical documentation.
September 2026	Integrated workflow development, platform readiness, and robotic manipulation. Revised the BPMN model based on project coordination meetings and evolving challenge requirements. Implemented automatic Spot navigation from the handoff station to storage, updated the QR-code identification scheme to use package-specific codes, and improved GraphNav mapping through loop closure. Prepared QR-coded task cards, package labels, operator control panels, and departure-hold mechanisms. Configured and tested the Unitree Go2W as a backup platform for the SUMMIT XL. Advanced development and testing of the FR3 Cartesian Dynamic Movement Primitive (DMP)-based pick-and-place system, incorporating wrist-mounted RGB-D perception, hand–eye calibration, grasp demonstration and teaching, and simulation-based validation.
October 2026	Final integration, testing, and documentation activities. Implemented the SUMMIT XL departure-hold control panel and consolidated the project software components and documentation into a unified repository. Continued FR3 hardware–software integration and end-to-end pick-and-place testing, including object-position adaptation, motion execution validation, debugging, and development of the operator GUI. Consolidated technical documentation and testing outcomes to support the assessment of project deliverables and preparation for the Robotic Awareness Event.

Development record

The SUMMIT XL, Spot, Go2W, arena simulation and BPMN work described in this document can be referenced from the project's GitHub repository: https://github.com/KinesisCTP/ad-ports-robotic-awareness/tree/main/edition-2. The repository holds the software for the mobile robots, the simulation, the BPMN source, the configuration and maps, the documentation, and the QR material print layouts. Every change is recorded as a commit, so the repository also shows how the work progressed.

The Franka FR3 pick-and-place implementation is documented separately in the NYUAD REACH Lab's private GitHub repository: https://github.com/NYUAD-REACH-LAB/franka_dmp_pick_place/tree/reachLab.

The commit history shows continuous development since June, with regular commits in every month from June to September, and work continuing into October (Figure 1). More than 100 commits were made during this period by the team members working on the project.

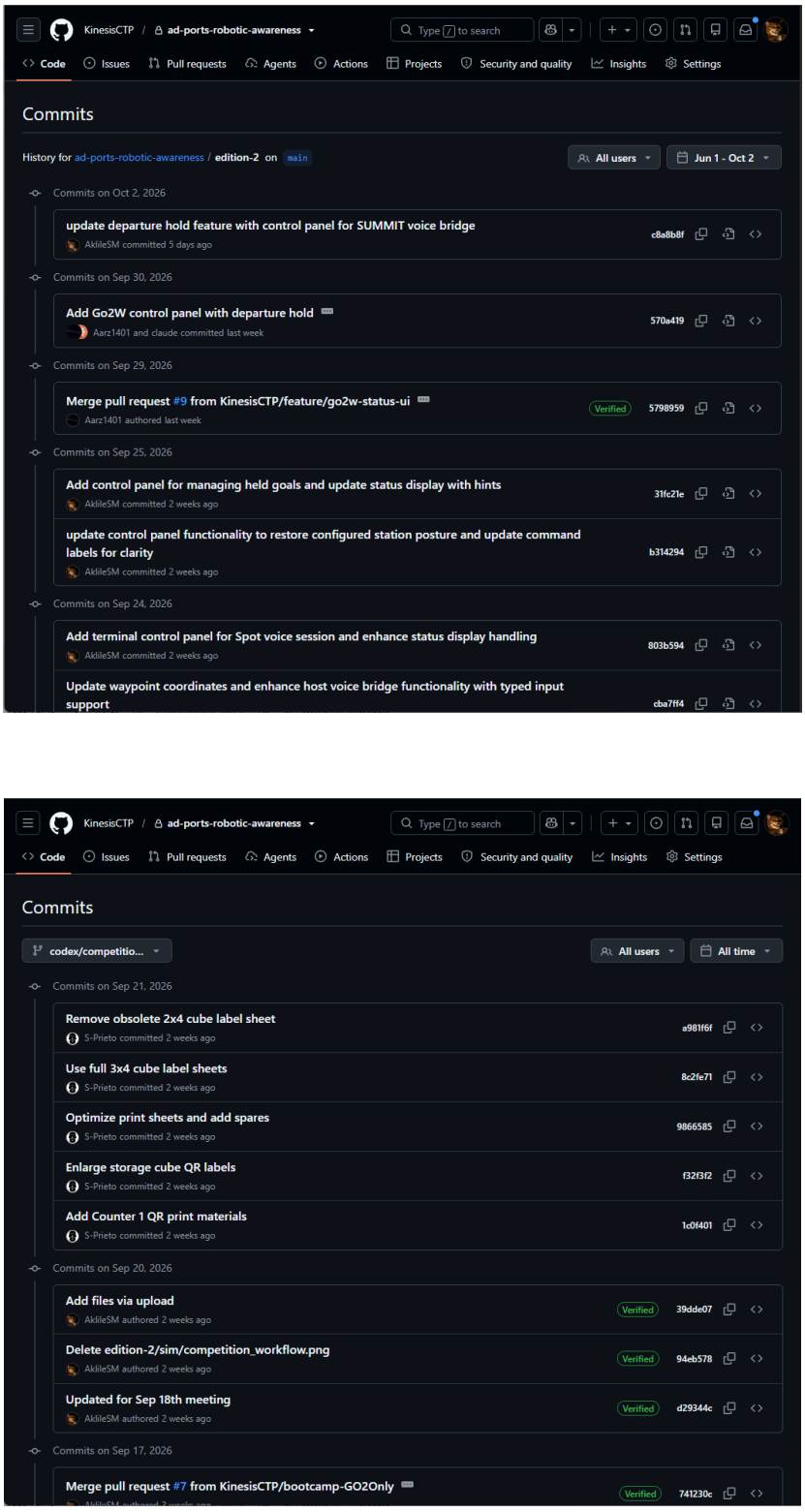


Figure 1. GitHub commit history samples showing regular commits from June to October.

3. Process Design and Workflow Modeling (BPMN)

The overall challenge workflow was modeled using Business Process Model and Notation (BPMN) 2.0, as illustrated in Figure 2. The model provides a structured representation of the operational sequence, responsibilities, decision points, and interactions between human operators and robotic platforms. Unlike a simulation script, the BPMN model captures the logical organization of the challenge, defining how tasks are initiated, how information and packages are transferred between stations, and how the different robotic systems coordinate to complete each delivery.

The process model incorporates nine swimlanes representing the participating entities: Organizer, P1 (Order Counter), SUMMIT XL, P2 (Handoff Station), P3 (Storage Station), Spot, P4 (Robotic Arm Station), FR3 Robotic Arm, and P5 (Delivery Station). This structure supports the identification of individual responsibilities, information exchanges, dependencies, and synchronization requirements across the challenge workflow.

The main operational sequence represented in the BPMN model is summarized as follows:

1. Task selection and initiation. Operator P1 selects a task corresponding to one of three difficulty levels (3, 6, or 12 points), requiring the delivery of one, two, or three color-coded packages, respectively. The Organizer issues the corresponding QR-coded task card, and the 15-minute challenge timer is initiated.
2. Task transfer to the handoff station. P1 places the QR-coded task card and the clicker on the SUMMIT XL tray and issues a navigation command directing the robot to the handoff station, where the task information is transferred to the next stage of the workflow.
3. QR-code scanning and task identification. At the handoff station, operator P2 presents the task QR code to Spot. Spot scans and processes the code to identify the required package colors and automatically navigates to the storage station.
4. Package identification and collection. At the storage station, operator P3 presents individual color-coded package QR codes to Spot for verification against the task requirements. When a matching package is identified, Spot confirms the match, and P3 places the corresponding package on the robot. P3 then issues a voice command directing Spot to the robotic arm station.
5. Robotic manipulation and package transfer. At the robotic arm station, operator P4 supervises and operates the FR3 robotic manipulator to retrieve the package from Spot and transfer it onto the SUMMIT XL. If additional packages are required to complete the selected task, Spot returns to the storage station, and the identification, collection, and transfer sequence is repeated until all required packages have been processed.
6. Final delivery and task completion. Once the required packages have been transferred, P4 commands the SUMMIT XL to navigate to the delivery station. Operator P5 receives the packages and places them in the designated delivery bracket to complete the task. If sufficient time remains within the 15-minute challenge period, P1 initiates another task, and the workflow repeats.

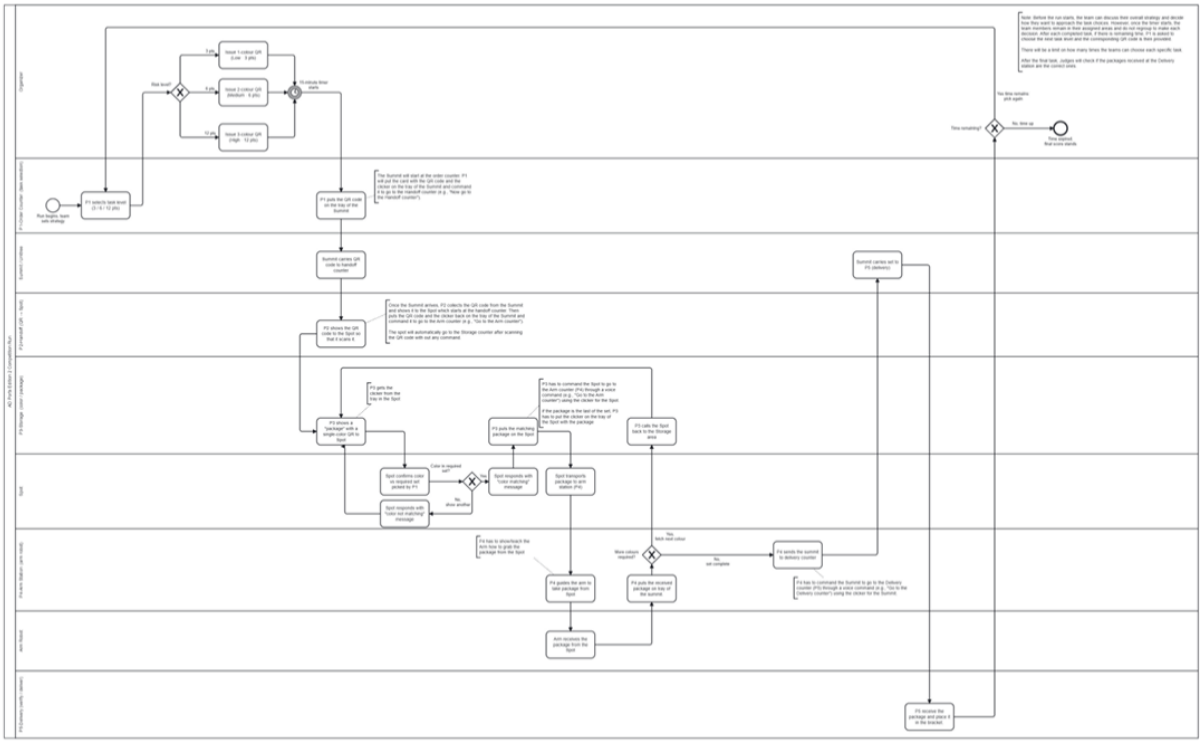


Figure 2. BPMN diagram with overview of process flow and key task..

The BPMN model was progressively reviewed and refined following project coordination meetings to incorporate changes in the challenge requirements and operational procedures. Key revisions included defining restrictions on the number of times each task difficulty level could be selected and clarifying the responsibilities of individual operators for commanding and interacting with each robotic platform. The workflows represented in the SUMMIT XL, Spot, and FR3 robotic arm swimlanes were translated into the corresponding software implementations described in the following sections, ensuring alignment between the process model and the developed robotic functionalities.

The model also explicitly distinguishes task execution from performance evaluation. Scoring and verification responsibilities remain exclusively with the Organizer and designated judges, who assess the packages received at the delivery station and assign points according to the established challenge rules. Participating teams and operators are responsible for executing the tasks but do not determine or award their own scores. This separation ensures consistency and impartiality in the evaluation process..

4. Simulation

A MuJoCo simulation of the arena was built to rehearse the workflow before using the robots (Figure 3). It models two separated zones joined only by a central arm station, with a handoff counter, storage rack, order counter and delivery counter, plus static operator models for P1 to P5.

- The scene is defined in one MuJoCo XML file and can be edited and reloaded live in the viewer.
- The run is a finite state machine that executes three work orders (packages A, B and X) through a 13-stage sequence: order issued, SUMMIT to handoff, Spot to storage, package onto Spot, Spot to arm station, arm takes the package, SUMMIT collects it, delivery.

- The wheeled SUMMIT XL replaced the earlier humanoid model in the simulation, matching the platform used in the real work. The arm is shown as a visual prop only; arm control was outside the scope of this arena simulation.
- Package colors map to delivery brackets, and the stage-by-stage progress prints to the terminal.

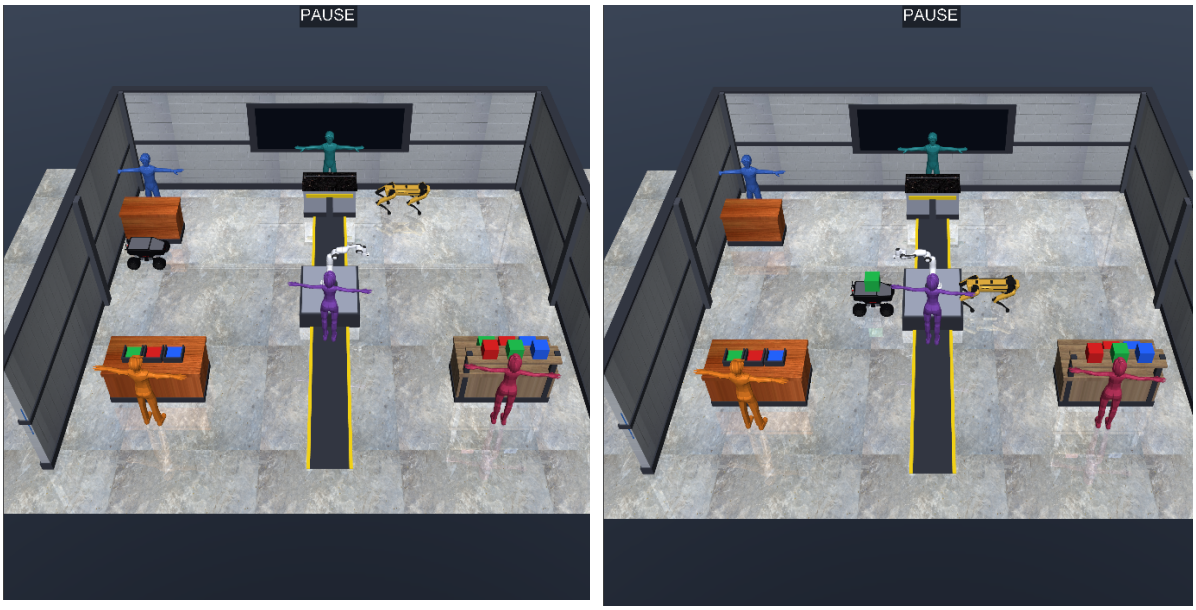


Figure 3. MuJoCo simulation of the Edition 2 arena.

A separate simulation environment was developed for the Franka Research 3 to test the manipulation workflow before running it on the physical robot (Figure 4). It reproduces the FR3, work surface, Franka Hand and package objects used for the pick-and-place task.

- The same Cartesian-DMP pick-and-place workflow used on the real FR3 can be run in simulation.
- Pick/place demonstrations, grasp behavior and changes in package position can be tested without using the physical robot.
- The simulator was used to check trajectory generation, object-position adaptation and end-to-end pick-and-place behavior before hardware trials.

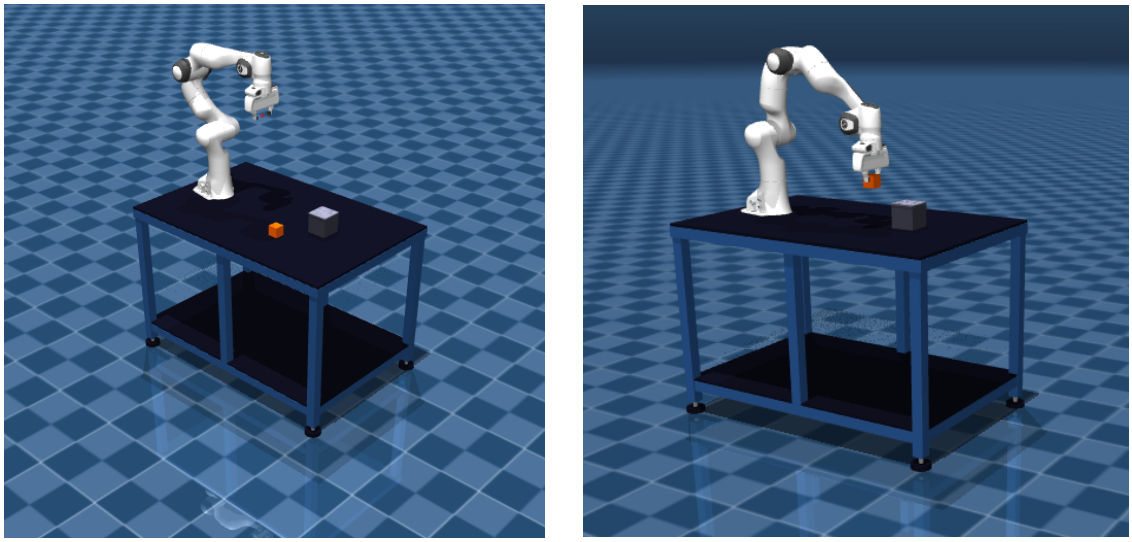


Figure 4. MuJoCo simulation of the FR3 pick-and-place task.

5. Robot platforms and software

5.1 Common design

The three mobile platforms share one operating idea, so an operator can use any of them the same way.

- **Voice or typed command.** The operator presses a USB clicker, speaks one destination and presses again. A wireless microphone records the speech, and OpenAI Whisper transcribes it locally. Every voice command can also be typed, but this feature is for debugging purposes only not included in the competition.
- **Closed-vocabulary matching.** The transcript is accepted only if it clearly names exactly one destination inside the arena. Word matching was used to achieve a fast voice-to-command sequence. On the SUMMIT, a command that the matcher cannot resolve is passed to a language model (LLM) for better understanding, and it is rejected if it still does not match a waypoint inside the arena. On Spot and Go2W, a command that does not match a waypoint inside the arena is rejected directly.
- **Safety.** Clicker presses are ignored while the robot is moving, so an accidental click cannot start or stop anything. The space bar sends a software cancel (Only for organizers and developers control). This is not an emergency stop: each robot's own e-stop remains the primary safety device.
- **Operator display.** A local full-screen browser page shows Initializing, Ready, Listening, Going to (destination) and Not a valid command, so someone standing at the robot can see what it is doing. On Spot the page also shows the QR-code stages (Waiting for QR code, Scanning QR code, Color code stored, QR code matched or not matched) and one small package icon for each stored code. Spot also gives audio feedback: a short voice prompt announces each QR result.
- **Control panel and departure hold.** A second terminal lets an overseer or developer release or cancel a goal at any time. The clicker and microphone travel on the robot's tray, so a goal to the handoff point is held for a short countdown timer, which gives the person at the counter time to load the tray before the robot leaves. The hold and release controls remain available for finer control during development and testing.

5.2 SUMMIT XL

The SUMMIT XL is a ROS 1 (Melodic) mecanum-wheel base (Figure 6). Sensing, localization and planning run on the robot (Ouster OS-1-64 lidar, Astra camera, PX4 IMU, AMCL and move_base with the TEB planner). A reusable Docker container on the operator laptop runs the map server, RViz and the command assistant, and a host-side bridge handles the clicker, microphone and speech recognition (Figure 5).

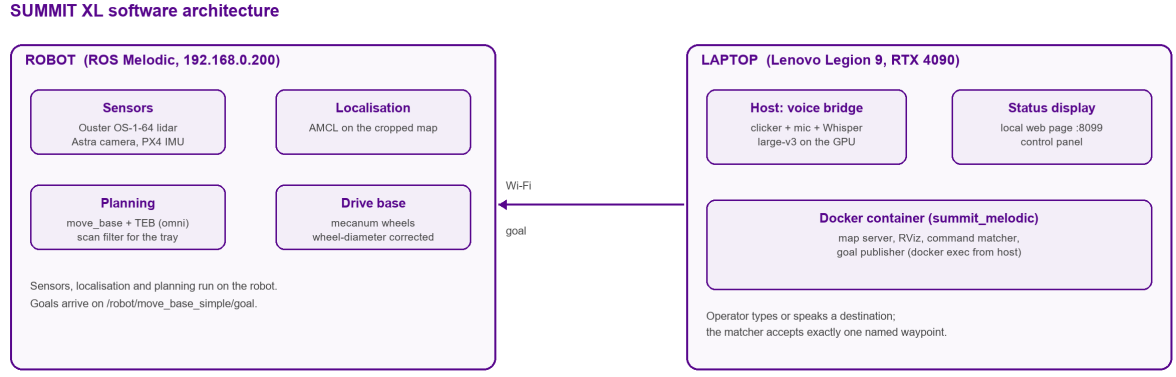


Figure 5. SUMMIT XL software architecture.

Work completed:

- Set up the reusable container with GPU-accelerated RViz, network routing to the robot, and the mapping workflow; built and cropped the arena map (about 3.9 million cells reduced to about 78,500) and recorded the four named waypoints: order counter, handoff, arm and delivery.
- Built the text assistant, then the voice and clicker bridge. Speech recognition uses Whisper large-v3 on the laptop GPU at roughly 0.16 to 0.47 s per command.
- Tuned navigation, including a wheel-diameter correction, localization and costmap parameters, and a footprint and scan filter for the rear tray.
- Added a navigation recovery watchdog, a speech-check tool that logs every attempt for later review without moving the robot, the status display (Figure 7), and the control panel with departure hold and timer.
- The departure hold counts down (10 s by default) and then departs on its own, or waits for the overseer if the countdown is disabled.



Figure 6. SUMMIT XL with the laser-cut tray and the mounted lidar.

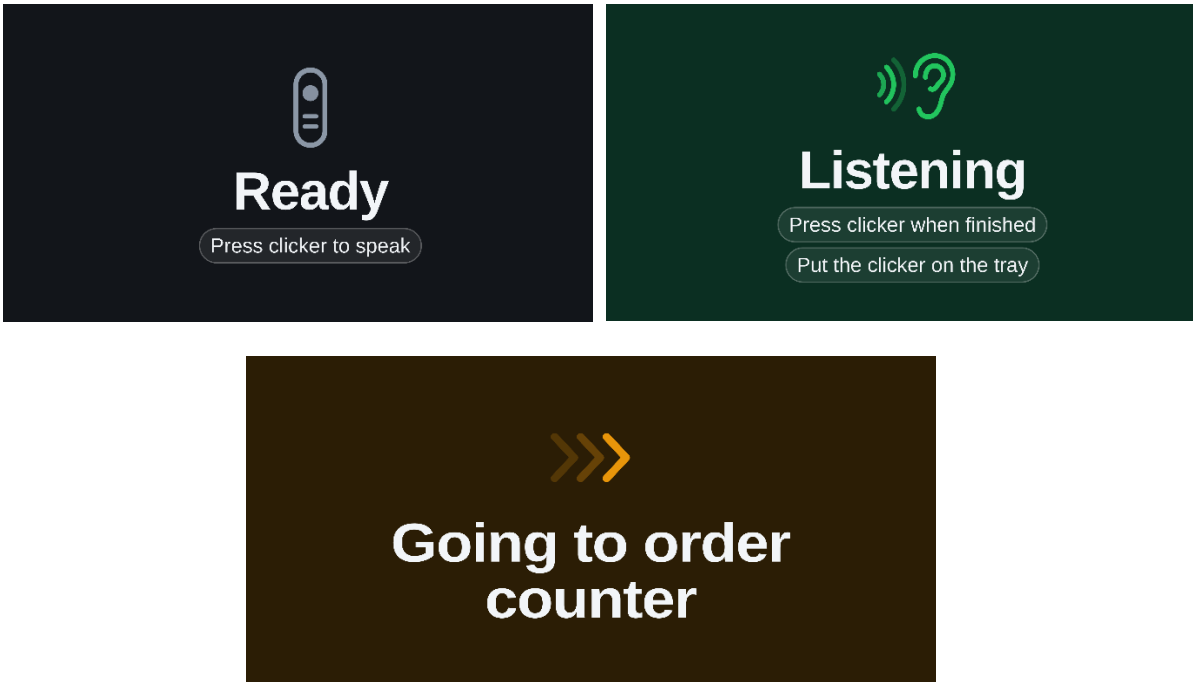


Figure 7. SUMMIT operator displays

5.3 Spot

Spot is controlled through the Boston Dynamics SDK from a NUC computer connected to the robot (Figure 8). Navigation uses GraphNav: a map of the arena was recorded and saved, and named waypoints are resolved against the active map. The software holds the robot's lease continuously and keeps it standing between commands.

Work completed:

- **Map workflow.** Record, back up, upload and localize a map from a fiducial; a topology step closes safe loops so Spot can take shorter routes.
- **Typed and voice navigation.** A parser normalizes natural phrasings such as "go to the handoff" or "take me to the arm", resolves each destination uniquely against the active map, and rejects vague, relative or chained commands. Whisper runs offline on the NUC, with the model cached beforehand because Spot's network has no internet.
- **QR scanning and matching.** After arriving at a configured waypoint, Spot scans with its front cameras. At the handoff it reads and stores one to three package codes (PKG-A to PKG-E) and then goes to storage automatically. At storage it reads a single package code and reports a match or no match; each match removes one entry from the stored list. QR text is never executed as a movement command.
- **Scanning posture.** A per-waypoint configuration tilts Spot's body so its cameras can read the codes, and it returns to neutral after a successful match.
- **Feedback.** The browser display shows Waiting for QR code, Scanning, Color code stored, QR code matched and QR code not matched, with one small package icon per stored entry. Cached voice prompts announce each outcome, timed to follow the screen (Figure 9).
- **Input handling.** Presses are ignored during navigation and QR scanning, and a terminal control panel can reset the state, re-arm scanning or cancel navigation.



Figure 8. Spot with the laser-cut tray fitted on its back.

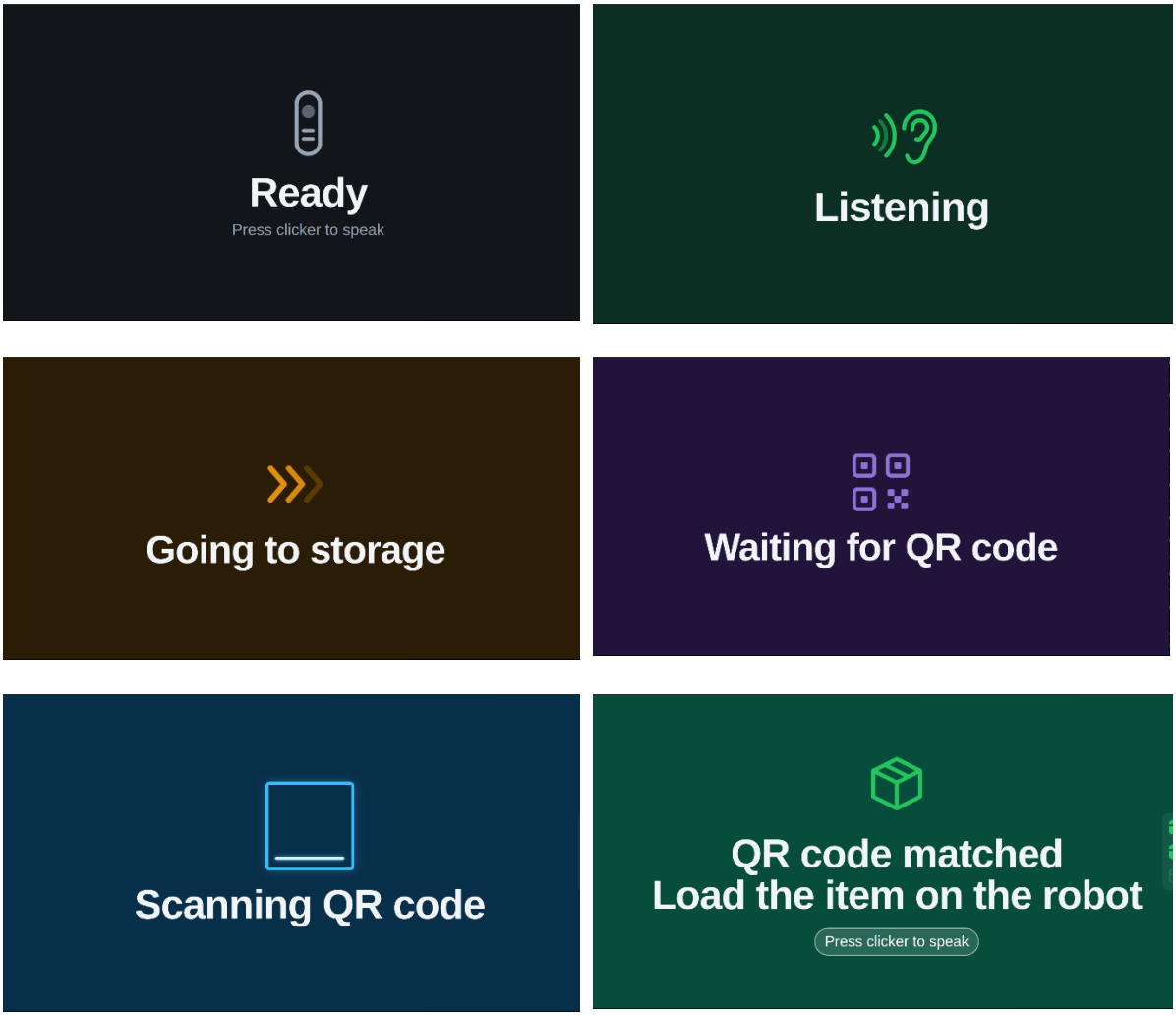


Figure 9. Spot operator display during QR handling.

5.4 Go2W (backup for the SUMMIT)

A Unitree Go2W (Figure 10) was prepared as a backup platform for the SUMMIT, using the same four destinations (order counter, handoff, arm, delivery) and the same operating flow. It runs ROS 2 with Nav2.

- Waypoints are read from RViz goal poses and stored in a single configuration file; the saved pose is sent directly to Nav2's navigate-to-pose action.
- Voice navigation uses the clicker, Whisper and the same closed-vocabulary matcher as the other platforms.
- A local status display and a terminal control panel (same as the SUMMIT) with departure hold were added (Aadil).
- A robot-free automated test suite (25 tests) covers the matcher, controls, display and waypoint handling.

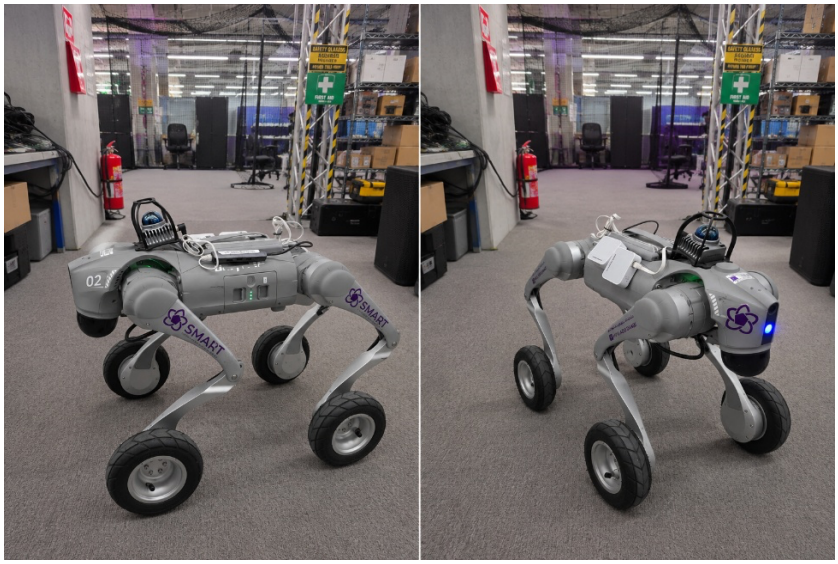


Figure 10. Go2W platform and operator display.

5.5 Franka Research 3

The Franka Research 3 (FR3) is the manipulation platform at the central arm station (Figure 11). A wrist-mounted Intel RealSense D455 provides RGB-D perception, while hand-eye calibration relates camera detections to the robot base frame. Pick and place motions are learned from hand-guided demonstrations using Cartesian Dynamic Movement Primitives (DMPs).

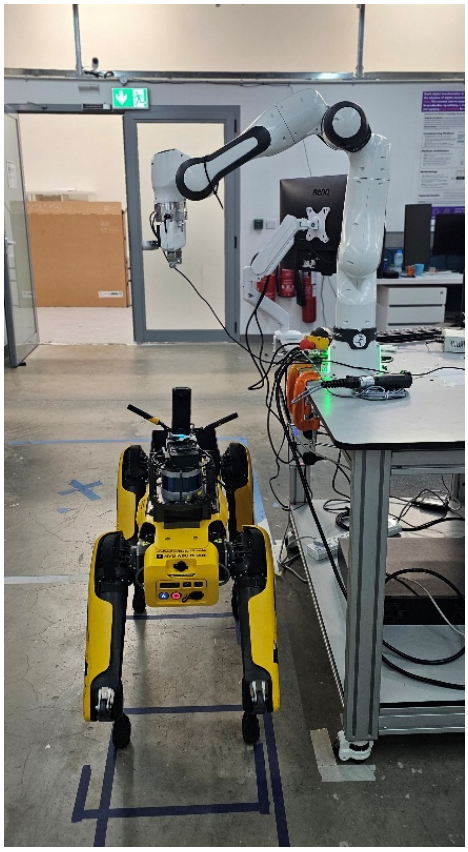


Figure 11. FR3 manipulation station with Spot positioned for package handoff.

Work completed:

- **Perception and calibration.** Developed wrist-camera blue-package detection, depth-based 3-D localization and hand-eye calibration.
- **Demonstration learning.** Recorded hand-guided pick and place trajectories and trained separate Cartesian DMPs for each motion.
- **Position adaptation.** Adapted the learned pick motion when the detected package position changes from the demonstrated position.
- **Grasp and motion validation.** Added grasp-width teaching, pointwise MoveIt inverse kinematics and trajectory checks before execution.
- **Operator interface.** Developed a GUI for setup, teaching, validation, execution and recovery of the FR3 workflow (Figure 12, Figure 13, and Figure 14).

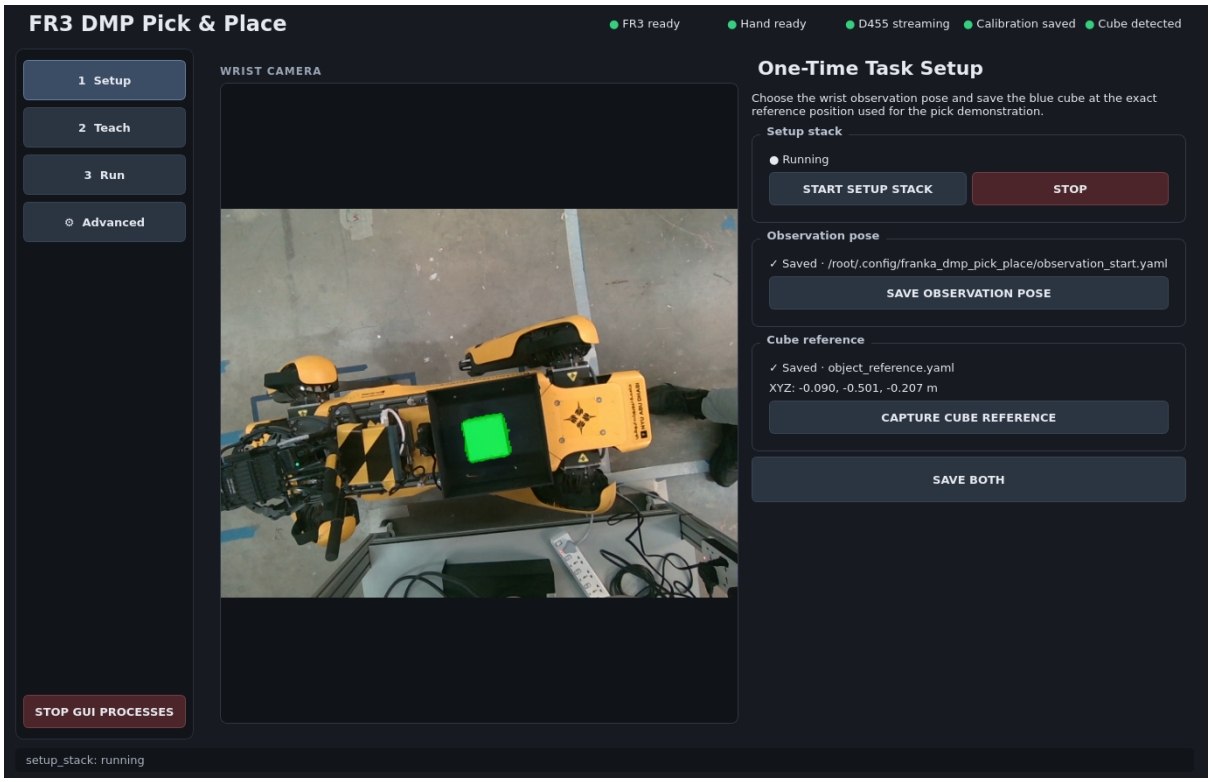


Figure 12. FR3 operator GUI for task setup and demonstration teaching.

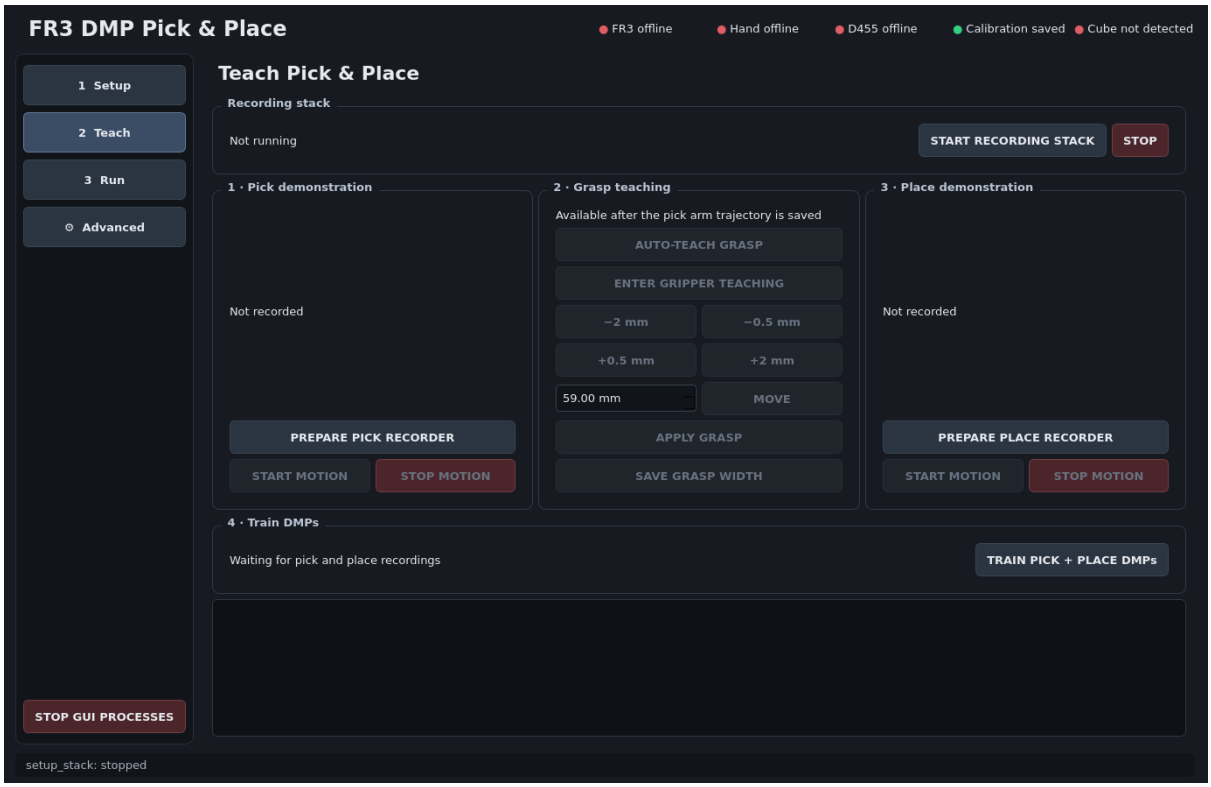


Figure 13. FR3 operator GUI for demonstration teaching.

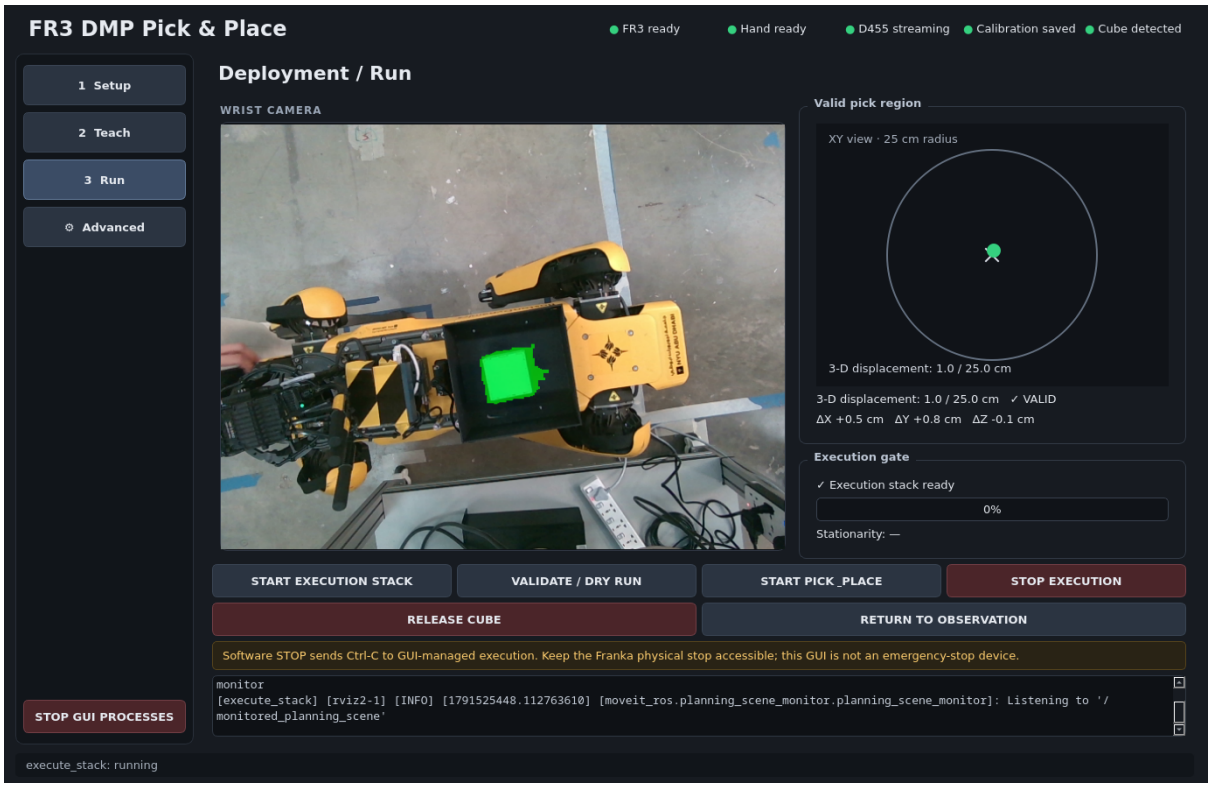


Figure 14. FR3 operator GUI for validation and pick-and-place execution.

6. Equipment, modifications and materials

6.1 Equipment used

Item	Use
SUMMIT XL mobile base (ROS Melodic)	Transport robot for the order, handoff, arm and delivery route. Ouster OS-1-64 lidar, Astra camera, PX4 IMU.
Boston Dynamics Spot	Shuttle between the handoff, storage and arm stations; scans QR codes with its own cameras.
Unitree Go2W	Backup for the SUMMIT (software prepared).
Franka Research 3 with Franka Hand	Manipulation platform at the arm station for package pick, transfer and placement.
Lenovo Legion 9 16IRX9 laptop, NVIDIA RTX 4090	Host for the SUMMIT: Docker container, map server, RViz, Whisper large-v3, voice bridge.
Intel NUC 11 computer	Connected to Spot: SDK control, offline Whisper, QR scanning, audio feedback.
Intel RealSense D455	Wrist-mounted RGB-D camera for package detection and 3-D localization.
Logitech clicker (USB receiver)(x2)	Start and stop voice recording. One per robot (SUMMIT and Spot), same model.
Hollyland wireless microphone and receiver set (x2)	Voice capture, same setup on SUMMIT and Spot. Travels on the robot's tray.
Spot tablet controller	Operator stop and e-stop during Spot testing.

Item	Use
Laser cutter	Cutting trays, item boxes and label parts.
Display for the status page	Operator display at the robot.
Software	ROS Melodic, ROS 2 / Nav2, MoveIt 2, Boston Dynamics Spot SDK, OpenAI Whisper, OpenCV, Docker, MuJoCo, PyQt5, Python.

6.2 Equipment modifications

The following equipment modifications and configuration activities were completed to support the integration, operation, and testing of the robotic systems:

- **Custom trays for SUMMIT and Spot**. Designed, fabricated, and installed custom trays on both robotic platforms to transport item boxes. The SUMMIT tray was fabricated from acrylic, while the Spot tray was fabricated from 3 mm-thick wood. The SUMMIT tray was dimensioned to accommodate the assembled boxes received from the robotic arm, providing sufficient space for their placement and transport.
- **SUMMIT navigation configuration**. Modified the robot's navigation footprint to account for the rearward extension of the installed tray. Additionally, implemented a LiDAR scan filter to prevent the robot's perception system from incorrectly identifying the tray as an obstacle, supporting reliable navigation with the modified configuration.
- **Spot configuration for QR code scanning**. Configured waypoint-specific body pitch adjustments to support QR code scanning during navigation. The corresponding parameters were defined and stored in a configuration file to enable consistent execution at designated waypoints.
- **Microphone and clicker integration**. Configured the microphone and clicker to be carried on the tray of the robot being commanded, supporting their use during robotic operation and testing.
- **FR3 wrist-mounted camera integration**. Designed and implemented a rigid mounting arrangement for the Intel RealSense D455 camera on the Franka Hand of the FR3 robotic arm. This modification enables wrist-mounted visual perception and supports the hand–eye calibration required for camera-based robotic manipulation.

These activities involved mechanical fabrication, hardware integration, sensor installation, and software configuration to adapt the robotic platforms to the project's operational requirements.

6.3 Materials

Material	Use
3 mm wood (laser cut)	Item boxes (cubes) and the Spot tray.
Acrylic (laser cut)	SUMMIT tray.
Printed QR task cards	Several sets with different codes (one to three package codes) for task selection.
Printed QR labels for item boxes	Labels matching the package codes (PKG-A to PKG-E), fixed to each box.
Printed ArUco calibration marker	Hand-eye calibration of the FR3 wrist-mounted camera.

The QR card and label sheets were prepared by Samuel, including print layouts for the counter work orders and the storage box labels. The trays and boxes were designed and cut together by Aklile and Samuel.



Figure 13. Samples of laser-cut item boxes with QR labels, and QR task cards.

7. Testing

7.1 Testing approach

- **Automated tests.** About 160 robot-free tests (135 for Spot, 25 for Go2W) cover command parsing, voice input, the QR scanner, session logic, the status displays and the control panels, so most behavior is checked without a robot. The FR3 package includes offline tests that do not connect to or command the robot.
- **Test modes without a robot.** The software can be run as a "dry run", which means it does everything except send commands to the robot. This covers a typed-only mode, a speech check that records, transcribes and reports what it would do, and a display preview that cycles through every screen. The speech check also logs each attempt as audio plus a record, building a labelled set from the real speakers and room. The FR3 workflow can also be tested in simulation, and its adapted pick-and-place trajectories can be validated through a dry run before execution.
- **Internal robot testing.** The SUMMIT, Spot, Go2W and FR3 were each tested internally on the full operating flow and worked as intended.

7.2 Operational notes

- The SUMMIT's time synchronization with the lidar is started by hand after each robot reboot, and the scan filter depends on it.
- Spot's speech model is downloaded and cached in advance because the robot network has no internet.
- The FR3 wrist camera must remain rigidly mounted after hand-eye calibration; recalibration is required if the mount changes.
- FR3 execution can be validated with a dry run from the Run tab of the operator GUI before real pick-and-place motion.

8. Project Team Roles and Technical Contributions

The project activities were carried out under the technical leadership and supervision of three Principal Investigators (PIs): Samuel A. Prieto, responsible for the Spot robotic platform, including its navigation, perception, hardware adaptations, and integration with the other systems; Fares Abu-Dakka, responsible for the FR3 robotic manipulation workstream; and Borja García de Soto, responsible for the SUMMIT XL and Go2W mobile robotic platforms, associated software development, and overall system integration. The project team’s contributions

encompassed system design, software development, hardware modifications, robotic integration, simulation, testing, and validation.

Person	Role and Technical Contributions
Fares Abu-Dakka (PI)	Principal Investigator responsible for the FR3 robotic manipulation workstream, including technical direction and supervision of the robotic arm's development and integration into the overall challenge. Oversaw the development of manipulation and motion-planning capabilities, wrist-mounted camera perception, hand–eye calibration, and automated pick-and-place operations. Provided technical guidance on robotic manipulation, integration, and validation to support reliable execution of the assigned tasks.
Samuel A. Prieto (CTP)	Principal Investigator responsible for the technical supervision and coordination of activities involving the Spot robotic platform. Oversaw and contributed to the definition of functional requirements, navigation and perception capabilities, hardware adaptations, and integration with the other robotic platforms. Provided technical guidance and hands-on co-development support throughout the project, including software integration, troubleshooting, and experimental testing. Prepared QR-coded task cards, identification labels, and other printed materials required for the challenge. Co-designed, fabricated, and assembled custom trays and item boxes for the robotic platforms. Developed the tutorial video documenting the system and its operation, and consolidated the project software repositories to support documentation, reproducibility, and continued development.
Borja García de Soto (PI)	Principal Investigator responsible for the technical direction, supervision, and coordination of activities involving the SUMMIT XL and Go2W robotic platforms and their integration into the overall challenge. Oversaw and contributed to system architecture, functional requirements, human–robot interaction, hardware adaptations, and integration across platforms. Provided technical guidance throughout development, reviewed implementation progress, and supervised system-level testing and validation to ensure alignment with the project objectives and deliverables.
Aklile Mengiste	Lead software developer for the SUMMIT XL, Spot, and Go2W robotic platforms. Designed and implemented functionalities covering autonomous navigation, voice-command and clicker-based interaction, QR code detection and processing, robot status visualization, and operator control interfaces. Developed and integrated simulation environments and BPMN-based task workflows to support coordinated robotic operations. Co-designed and laser-cut custom trays and item boxes for physical task execution. Conducted iterative integration, troubleshooting, and internal testing across the robotic platforms.
Aadil Chasmawala	Provided support for the implementation and integration of the Go2W robotic platform, developing and configuring the software components required for its participation in the challenge. Worked on adapting and implementing the platform-specific functionalities needed to support the overall robotic workflow. Contributed to troubleshooting, iterative refinement, system integration, and testing throughout the development process, working alongside the broader software development team.
Aman Sunesh	Developed the FR3 operator graphical user interface (GUI) and carried out hardware–software integration and experimental testing on the physical robotic arm. Implemented and validated hand–eye calibration, wrist-mounted RealSense camera perception, and automated grasp control

	functionalities. Conducted dry-run validation, troubleshooting, and end-to-end pick-and-place trials to assess the integration and operation of the robotic manipulation system within the challenge workflow.
Omar Mostafa	Developed the ROS 2 implementation of Cartesian Dynamic Movement Primitives (DMPs) for FR3 pick-and-place manipulation. Implemented demonstration recording and DMP training, object-position adaptation for task execution, and inverse kinematics (IK) validation using MoveIt. Developed the corresponding robotic simulation twin to support motion-planning development, algorithm testing, and validation before deployment on the physical system.

These contributions supported the progressive development and integration of the robotic systems, from initial design and configuration to implementation, testing, and validation. The activities collectively contributed to the technical deliverables associated with autonomous mobile robot operation, human-robot interaction, robotic manipulation, and execution of the integrated challenge workflows.

In addition to their technical supervision and development responsibilities, the PIs participated in weekly Robotics Challenge v2 – Coordination Meetings with Joan Esteba and Pablo Brizzio (AD Ports). This regular coordination meetings helped maintain alignment among the project stakeholders, facilitate technical decision-making, and support the timely execution of project activities